

```
'boot=UUID=<your_uuid>'
```

The configuration files contain the live USB lines (commented). So, comment the lines for the live CD and uncomment the lines for the live USB to get the desired changes.

Phase V: Making the ISO image

Recheck the *mbcd/* for errors—check if all the paths to the configuration files are valid, whether all the files are copied, unnecessary back-up files ending with tilde have been removed “~”, etc.

We will use *mkisofs* to make the ISO image. Run the following command if you are making a Grub bootable image.

```
$ mkisofs -iso-level 3 -R -J --input-charset=utf8 \
  -b boot/grub/stage2_eltorito -c boot/boot.catalog \
  -no-emul-boot -boot-load-size 4 -boot-info-table -o mbcd.iso mbcd
```

If you want to make the ISO image boot with ISOLINUX instead, point to the *isolinux.bin* boot image:

```
$ mkisofs -iso-level 3 -R -J --input-charset=utf8 \
  -b boot/isolinux/isolinux.bin -c boot/boot.catalog \
  -no-emul-boot -boot-load-size 4 -boot-info-table -o mbcd.iso mbcd
```

Now burn it into a CD-R.



Tip: Test the ISO file in a virtual machine before you burn a CD.

Phase VI: Making the Swiss knife

Now it's time to convert your USB Flash drive to a GNU/Linux-powered, home-made Swiss knife with seven flavours.

We will describe how to make your USB drive bootable with both SYSLINUX and Grub. Though this process is not destructive, it is always advisable to back up data (see ‘Notes on USB Backup’). First copy all the contents of *work_dir/mbcd/* into the USB drive partition:

```
$ cp -r mbcd/* /media/disk/
```

Then install the Grub or SYSLINUX bootloader and set the boot flag.

[1] Installing SYSLINUX

In the USB drive, rename the directory *isolinux/* to *syslinux/*, rename *isolinux.cfg* to *syslinux.cfg* and *isolinux.bin* to *syslinux.bin*. Assuming your USB drive is mounted in */media/disk*, execute:

```
$ mv /media/disk/boot/isolinux /media/disk/boot/syslinux
$ mv /media/disk/boot/syslinux/isolinux.cfg /media/disk/boot/syslinux/
  syslinux.cfg
$ mv /media/disk/boot/syslinux/isolinux.bin /media/disk/boot/syslinux/
  syslinux.bin
```

Now install SYSLINUX in the USB drive's partition where you have the boot directory:

```
# syslinux -sd boot/syslinux /dev/sdXy
```

If the USB drive partition that has the bootloader is */dev/sdb1*, then the above command would be:

```
# syslinux -ds boot/syslinux/ /dev/sdb1
```

This will alter the MBR (Master Boot Record) by installing the boot code and installing *ldlinux.sys* in the partition.

[2] Installing Grub

The *grub-install* shell script will be used to do this job. The following command installs Grub in both the MBR and the partition.

```
# grub-install --root-directory=/media/disk /dev/sdX
```

The */media/disk/* directory should point to the mount point of your USB partition. This will install the boot code in the MBR and install Stage1.5 and Stage 2 into the partition and set the configuration file.



Note: Use the Grub shell for better control.

[3] Set the boot flag

Unmount the USB partition and make it bootable by setting the boot flag with *parted*, *fdisk* or any partition manager.

```
# umount /dev/sdXy
# parted /dev/sdX set y boot on
```

For example, if your USB drive partition is */dev/sdb1* (containing the boot directory), then the above commands would be:

```
# umount /dev/sdb1
# parted /dev/sdb set 1 boot on
```

After you have installed either Grub or SYSLINUX, make the changes in the configuration file as described in Phase IV, Section [3].